

EMDA and CFD in Numerical Relativity

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ABSTRACT

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[The abstract is a summary of the thesis/dissertation with emphasis on the findings of the study. The abstract must not exceed 350 words in length and fit on one page, single spaced.]

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Chapter 1

Introduction

1.1 Introduction

Since Einstein introduced the theory of general relativity in 1915 [1], it has remained consistent with all experimental tests to date. General relativity has fundamentally transformed our understanding of space and time, explaining phenomena ranging from the anomalous precession of Mercury to the operation of atomic clocks in GPS satellites, and predicting the existence of compact objects such as neutron stars and black holes, as well as gravitational waves [2, 3].

Gravitational waves were first directly detected by LIGO in 2015 [4]. These signals originate from some of the most energetic events in the universe, such as mergers of neutron stars and black holes. However, the signals detected by LIGO are extremely weak; the detectors are sensitive enough to measure relative length changes on the order of 10^{-21} , corresponding to displacements smaller than a proton diameter [5]. Despite this extraordinary sensitivity, gravitational wave signals remain weak relative to background noise. The highest signal-to-noise ratio (SNR) observed to date is on the order of ~ 80 [6].

Interpreting gravitational wave detections requires accurate theoretical models of the expected signals. In practice, this is achieved by comparing observational data to waveform templates generated through numerical simulations. This comparison allows us to statistically determine key physical properties of the source, such as masses and spins.

Making theoretical models of binary black hole mergers is not trivial. In GR, analytic solutions in general relativity are typically restricted to highly symmetric systems and cannot describe the full dynamics of binary black hole mergers. As a result, numerical relativity simulations are essential. A number of codes have been developed for this purpose, including the Einstein Toolkit, SpEC, GRChombo, BAM, GR-Athena++, and Dendro-GR. [7–12]

Simulating binary black hole mergers is computationally demanding. It requires solving a coupled system of nonlinear partial differential equations across multiple length scales, including

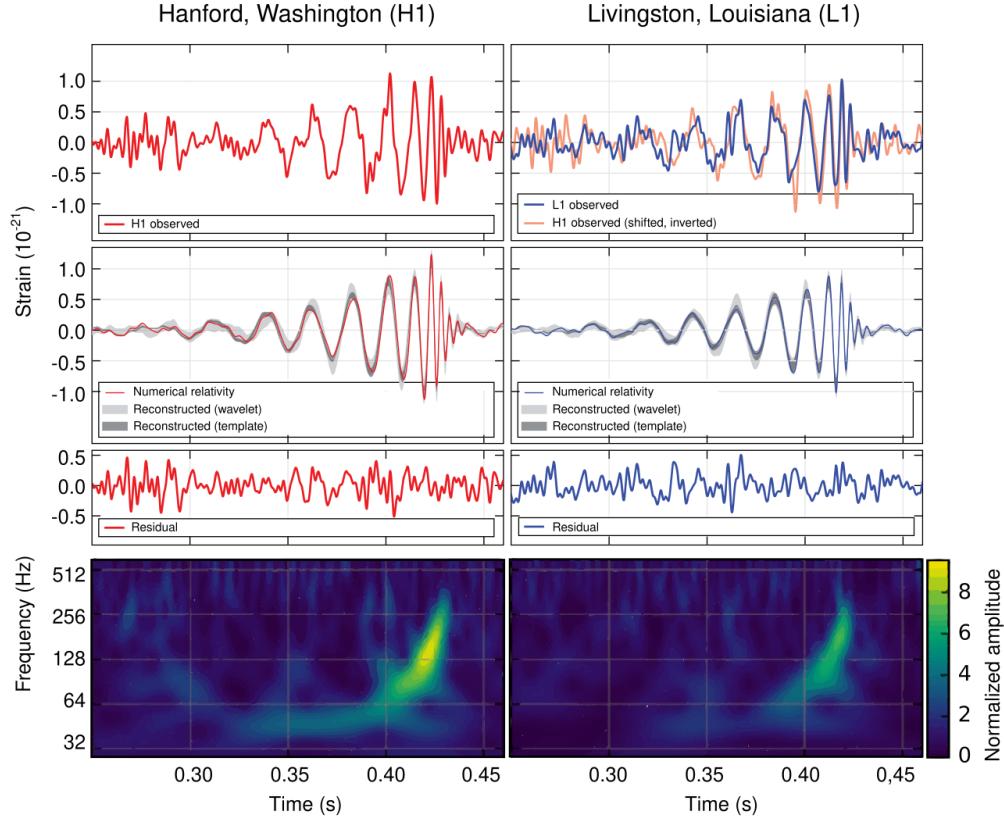


Figure 1.1 Schematic illustration of the LIGO interferometer used to detect gravitational waves through differential arm length changes.

the scale of the black holes, the binary orbital scale, and the radiation zone [13, 14]. In this work, we make use of the Dendro numerical relativity framework [12].

For standard general relativity (GR), substantial progress has been made, and extensive waveform catalogs now cover much of the parameter space, particularly for near-equal-mass systems (We here define the mass ratio q as $q = \frac{M_{\text{larger}}}{M_{\text{smaller}}} \geq 1$). The LIGO–Virgo–KAGRA (LVK) collaboration has reported hundreds of detected events using these models [15]. However, two key challenges remain: improving the fidelity of GR simulations and developing waveform catalogs for alternative theories of gravity .

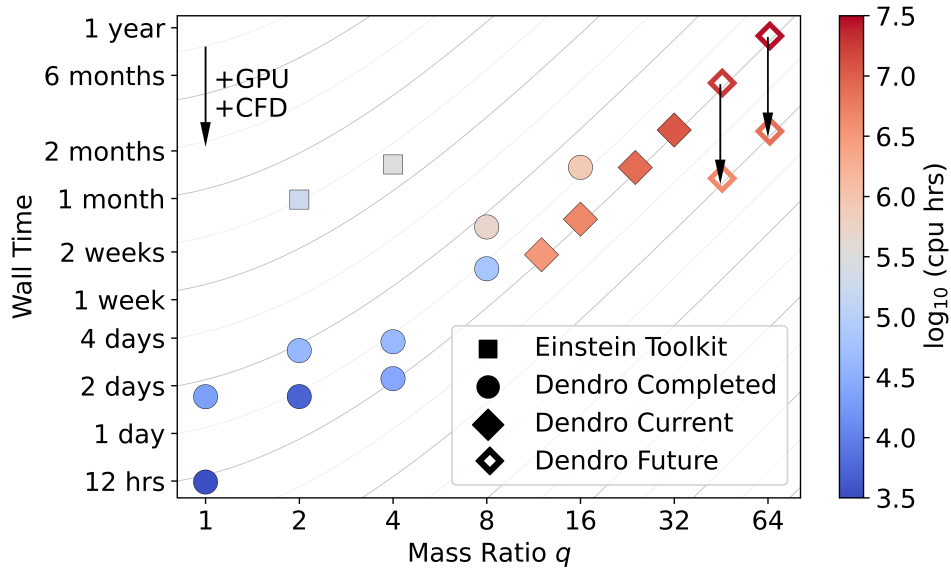


Figure 1.2 Walltime of numerical relativity simulations as a function of mass ratio.

The next generation of gravitational wave detectors is expected to significantly improve the signal-to-noise ratio of observed signals. Meeting this challenge will require higher-fidelity numerical simulations and more accurate waveform models.

In general, higher-fidelity simulations—particularly those with larger mass ratios—require finer grid resolution, leading to a substantial increase in computational cost. This trend is illustrated in Fig. 1.2. Current simulations with mass ratios $q < 8$ can require weeks of runtime on modern supercomputers, and further increases in resolution would significantly amplify this cost.

To obtain more accurate waveforms, there are two main approaches: running existing codes at higher resolution or improving the underlying numerical methods. A drawback of the brute-force approach—simply increasing resolution—is that the timescale of the problem is constrained by the CFL factor. This, in turn, requires more timesteps, significantly increasing the total runtime of the simulation.

One potential avenue for improving efficiency in numerical relativity codes lies in the computation of spatial derivatives. Most current implementations rely on explicit finite difference methods, while

spectral methods—although highly accurate—are often difficult to implement. In this work, we explore compact finite difference (CFD) operators as an alternative. Compact schemes are implicit methods that can achieve higher spectral resolution than comparable explicit schemes [16]. In tests within our Maxwell solver, CFDs often perform comparably to explicit methods at approximately half the resolution. Despite their success in computational fluid dynamics and related fields, their use in numerical relativity remains limited [17].

Gravitational wave observations provide a powerful probe of gravity in the strongly nonlinear, dynamical regime, where deviations from general relativity (GR) may become apparent. To search for such deviations, waveform templates from alternative theories of gravity are required. While some work has been carried out in theories such as Chern–Simons gravity, scalar–tensor theories, Einstein–Gauss–Bonnet gravity, and Einstein–Maxwell–dilaton theory, waveform catalogs in these frameworks remain incomplete.

In this dissertation, we focus on Einstein–Maxwell–dilaton–axion (EMDA) theory. EMDA is of particular interest because it admits a well-posed initial value formulation and arises as the low-energy limit of string theory. In contrast to the no-hair theorem in standard GR, which states the black holes can be completely defined by their mass spin and charge, EMDA black holes possess additional degrees of freedom associated with the dilaton and axion fields.

We present a numerical relativity framework for simulating binary black hole mergers in EMDA theory. We analyze the resulting waveforms and investigate whether deviations from GR may be observable in gravitational wave signals.

1.2 Background

We focus on the study of black hole dynamics. As such, we begin with Einstein’s field equations:

$$G_{ab} = 8\pi T_{ab} + \Lambda g_{ab}. \quad (1.1)$$

The left-hand side describes the geometry of spacetime, while the right-hand side encodes the matter and energy content. [18, 19] We briefly examine each term in turn.

The tensor G_{ab} is the Einstein tensor, which is constructed from quantities that encode the curvature of spacetime. The stress-energy tensor T_{ab} describes the distribution of matter and energy. The metric tensor g_{ab} defines distances and causal structure in spacetime. The constant Λ is the cosmological constant, which is generally assumed to be very small. On the scales considered in this work, it is negligible, and we therefore set $\Lambda = 0$.

In standard general relativity simulations of binary black hole mergers, we consider vacuum spacetimes, for which

$$G_{ab} = 0. \quad (1.2)$$

This provides the system of partial differential equations that must be solved in vacuum GR. Before proceeding further, we review some essential geometric concepts.

We begin with the metric, which encodes the notion of separation in spacetime. As an example, the Minkowski metric describing flat spacetime can be written as

$$ds^2 = -dt^2 + dx^2 + dy^2 + dz^2, \quad (1.3)$$

or equivalently in matrix form as

$$\eta_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (1.4)$$

Here we use η_{ab} to denote the flat spacetime metric. The negative sign in the time component reflects our choice of signature $(-, +, +, +)$.

The Minkowski metric can also be expressed in different coordinate systems. For example, in spherical coordinates it takes the form

$$\eta_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2 \theta \end{pmatrix}. \quad (1.5)$$

Although these representations appear different, they describe the same flat spacetime. This illustrates that the form of the metric depends on the choice of coordinates.

A natural question is how to determine whether a given metric describes flat spacetime. A necessary and sufficient condition is that the Riemann curvature tensor vanishes everywhere. Given a metric, the Riemann tensor is defined as

$$R^a{}_{bcd} = \partial_c \Gamma^a{}_{bd} - \partial_d \Gamma^a{}_{bc} + \Gamma^a{}_{ec} \Gamma^e{}_{bd} - \Gamma^a{}_{ed} \Gamma^e{}_{bc}, \quad (1.6)$$

where $\Gamma^a{}_{bc}$ are the Christoffel symbols, given by

$$\Gamma^a{}_{bc} = \frac{1}{2} g^{ad} (\partial_b g_{dc} + \partial_c g_{bd} - \partial_d g_{bc}). \quad (1.7)$$

These quantities encode the curvature of spacetime.

Contracting the Riemann tensor yields the Ricci tensor,

$$R_{ab} = g^{cd} R_{acbd}, \quad (1.8)$$

and further contraction gives the Ricci scalar,

$$R = g^{ab} R_{ab}. \quad (1.9)$$

The Einstein tensor is then constructed as

$$G_{ab} = R_{ab} - \frac{1}{2} g_{ab} R, \quad (1.10)$$

which satisfies the contracted Bianchi identities and ensures local conservation of energy and momentum.

With the Einstein tensor in hand, we can rewrite the field equations as a set of coupled second-order partial differential equations. To evolve these equations numerically, we must decompose them into a form suitable for time evolution. This requires splitting spacetime into three spatial dimensions and one time dimension.

The most straightforward approach leads to the ADM (Arnowitt–Deser–Misner) formulation. However, the ADM system is only weakly hyperbolic and can become unstable over long evolutions. Several modified formulations have been developed to address this issue. In this work, we use the BSSN (Baumgarte–Shapiro–Shibata–Nakamura) formulation, which is strongly hyperbolic and can be evolved stably for long periods of time. Given this formulation, we require suitable initial data, after which the system can be evolved forward in time.

To provide an overview of this procedure, we briefly consider simpler examples: a scalar field and Maxwell’s equations.

We begin with the scalar wave equation with a source term:

$$\square\psi = \nabla_a \nabla^a \psi = 4\pi\rho. \quad (1.11)$$

This equation is written in fully covariant form. Expanding it in flat spacetime using the Minkowski metric, we obtain

$$\square\psi = \eta^{ab} \nabla_a \nabla_b \psi = \left(-\partial_t^2 + D^2 \right) \psi, \quad (1.12)$$

where D denotes purely spatial derivatives.

This provides a simple example of a 3+1 decomposition. The scalar wave equation is a second-order hyperbolic equation with well-posed initial value properties: solutions exist and are unique. Importantly, there are no constraint equations in this system, meaning we are free to choose ψ and ρ arbitrarily as initial data. As we will see, this is not the case for Maxwell’s equations or general relativity, where constraint equations must be satisfied.

We now turn to the more complex case of Maxwell's equations:

$$\nabla \cdot \mathbf{E} = 4\pi\rho, \quad (1.13)$$

$$\nabla \cdot \mathbf{B} = 0, \quad (1.14)$$

$$\frac{\partial \mathbf{E}}{\partial t} = \nabla \times \mathbf{B} - 4\pi\mathbf{J}, \quad (1.15)$$

$$\frac{\partial \mathbf{B}}{\partial t} = -\nabla \times \mathbf{E}. \quad (1.16)$$

We present the full 3 + 1 decomposition of Maxwell's equations in the appendix. However, an important feature is already evident: there are eight equations, but only six dynamical variables $(E_x, E_y, E_z, B_x, B_y, B_z)$. The remaining two equations act as constraints on the system.

Maxwell's equations can be separated into evolution equations:

$$\frac{\partial \mathbf{E}}{\partial t} = \nabla \times \mathbf{B} - 4\pi\mathbf{J}, \quad (1.17)$$

$$\frac{\partial \mathbf{B}}{\partial t} = -\nabla \times \mathbf{E}, \quad (1.18)$$

and constraint equations:

$$\nabla \cdot \mathbf{E} = 4\pi\rho, \quad (1.19)$$

$$\nabla \cdot \mathbf{B} = 0. \quad (1.20)$$

Specifically, Gauss's law and the absence of magnetic monopoles must be satisfied at all times. If the initial data satisfy these constraints, then in the continuum limit the evolution equations will preserve them. However, in numerical simulations we work with finite precision, and violations of the constraint equations provide a useful measure of numerical error.

Solving the Cauchy problem for Einstein's equations is nontrivial, as we must satisfy both the Hamiltonian and momentum constraints. There are established techniques for solving these constraints. In particular, the momentum constraints are often addressed using the Bowen–York formalism. The Hamiltonian constraint reduces to an elliptic equation that must be solved numerically.

In EMDA, solving the Cauchy problem is further complicated by the presence of additional source terms. The stress-energy tensor T_{ab} encodes all contributions from matter and energy apart from gravity. In standard GR this is 0, in EMDA, this includes electromagnetic fields as well as scalar fields. The solution for single black holes is known analytically for the certain coupling strengths ($\alpha_0 = \alpha_1 = 1$) but numerically one could also imagine varying the coupling strength to see how the evolutions change. This dissertation will address the challenges associated with constructing consistent initial data and evolving these systems.

Once we have constructed suitable initial data and evolved the system, we obtain a gravitational waveform. We compute this waveform using the Newman–Penrose formalism, in which it is represented by the scalar Ψ_4 . Details of how Ψ_4 is calculated are provided in the appendix.

From Ψ_4 , we can reconstruct the gravitational-wave strain measured by LIGO. This allows us to compare our simulated waveforms with observed signals. To quantify the difference between two waveforms, we compute the mismatch, defined as

$$\text{mismatch}(h_1, h_2) = 1 - \max O(h_1, h_2), \quad (1.21)$$

where the maximum is taken over time shifts and polarization angles.

From the mismatch, we can estimate the minimum signal-to-noise ratio (SNR) required to distinguish between the two waveforms. This is given by

$$\text{SNR}(h_1, h_2) = \frac{1}{\sqrt{2 \text{mismatch}(h_1, h_2)}}. \quad (1.22)$$

Despite this framework, interpreting differences between waveforms remains challenging. The parameter space is often degenerate; for example, it may not be immediately clear whether observed differences arise from spin, charge, or other physical effects.

We additionally present the work we have done incorporating compact finite differences into our numerical relativity code. To begin, we first discuss the methods currently used in the code.

On a discrete grid, we can approximate derivatives using finite differences. For example, a second-order accurate centered approximation to the first derivative is given by

$$\left. \frac{df}{dx} \right|_i = \frac{f_{i+1} - f_{i-1}}{2\Delta x} + O(\Delta x^2). \quad (1.23)$$

For higher-order accuracy, we can use a Taylor series expansion to construct more accurate approximations. For instance, a fourth-order accurate stencil is

$$\left. \frac{df}{dx} \right|_i = \frac{-f_{i+2} + 8f_{i+1} - 8f_{i-1} + f_{i-2}}{12\Delta x} + O(\Delta x^4). \quad (1.24)$$

These are examples of explicit finite difference schemes. Alternatively, we can seek an approximation to the first derivative using a compact (implicit) scheme of the form

$$\beta f'_{i-2} + \alpha f'_{i-1} + f'_i + \alpha f'_{i+1} + \beta f'_{i+2} = c \frac{f_{i+3} - f_{i-3}}{6\Delta x} + b \frac{f_{i+2} - f_{i-2}}{4\Delta x} + a \frac{f_{i+1} - f_{i-1}}{2\Delta x}. \quad (1.25)$$

The coefficients can be determined by matching Taylor series expansions to the desired order of accuracy (a more detailed derivation is provided in the appendix):

$$a + b + c = 1 + 2\alpha + 2\beta \quad (2\text{nd order}), \quad (1.26)$$

$$a + 2^2b + 3^2c = 2 \frac{3!}{2!} \left(\alpha + 2^2\beta \right) \quad (4\text{th order}), \quad (1.27)$$

$$a + 2^4b + 3^4c = 2 \frac{5!}{4!} \left(\alpha + 2^4\beta \right) \quad (6\text{th order}). \quad (1.28)$$

This approach is more complex and leads to an implicit scheme. However, the trade-off is improved accuracy and enhanced spectral resolution compared to explicit methods. This improvement in spectral resolution is illustrated in Fig. 1.3.

There are many possible compact finite difference operators that can be constructed. In this work, we describe our methodology for identifying accurate and stable compact derivative operators and evaluate their performance relative to their explicit counterparts.

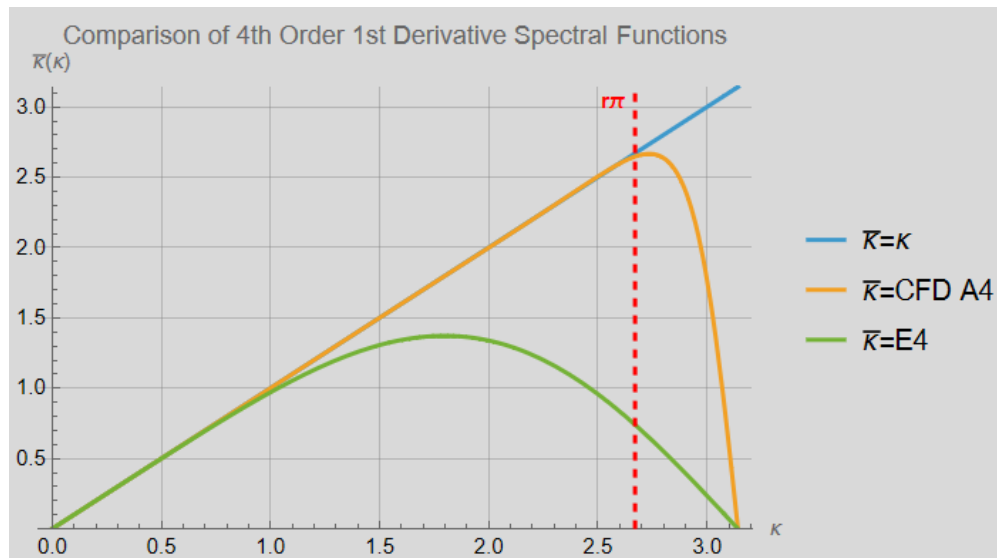


Figure 1.3 Comparison of spectral resolution for explicit and compact fourth-order first-derivative operators.

1.3 Summary of Dissertation

In this dissertation, we primarily present results from binary black hole simulations in EMDA theory. The dissertation is organized as follows. Chapter 2 discusses the formulation of the initial data and presents results from head-on collision simulations. Chapter 3 presents inspiral and merger results in EMDA theory. Chapter 4 explores interpolated EMDA models in which the coupling strengths are varied. Chapter 5 investigates the implementation of compact finite difference operators in numerical relativity codes.

The results presented in this dissertation are currently in preparation for publication.

Each of these projects involved multiple collaborators. The EMDA work builds on earlier contributions by Hyun Lim (LANL) and was carried out collaboratively with Eric Hirschmann (BYU), David van Komen (University of Utah), Sebastian Vander Ploeg Fallon (Cornell), and David Neilsen (BYU). The compact finite difference (CFD) project was conducted in collaboration with

Eric Hirschmann (BYU), David van Komen (University of Utah), David Neilsen (BYU), as well as Nathaneil Garey and Luke Papenfuss (BYU).

1.4 Conventions

In this work, we adopt geometrized units such that

- $c = G = 1$.
- Lengths and masses are expressed in units of a characteristic mass scale M .
- Distance: $d = M \Rightarrow d = GM/c^2$.
- Time: $t = M \Rightarrow t = GM/c^3$.
- We use the metric signature $(-, +, +, +)$.
- Einstein summation convention is assumed unless otherwise stated.
- Spacetime indices $a, b, c, \dots$ run over $0, 1, 2, 3$, while spatial indices $i, j, k, \dots$ run over $1, 2, 3$.
- The symmetric and antisymmetric parts of a tensor are defined as

$$T_{(ab)} \equiv \frac{1}{2} (T_{ab} + T_{ba}), \quad T_{[ab]} \equiv \frac{1}{2} (T_{ab} - T_{ba}).$$

- Partial derivatives are denoted by $\partial_i g$ or equivalently $g_{,i}$.
- Covariant derivatives are written as $\nabla_a f_b$ or equivalently $f_{b;a}$.
- The Lie derivative of a tensor X^a along a vector field β^a is denoted by $\mathcal{L}_\beta X^a$.
- In general, we refer to objects associated with

– the spacetime manifold $\mathcal{M}$ as g_{ab} , ${}^{(4)}\Gamma^a_{bc}$, ∇_a , ${}^{(4)}R_{ab}$, etc.,

- a spatial slice Σ as γ_{ij} , Γ^i_{jk} , D_i , R_{ij} , etc.,
- a conformally related space as $\bar{\gamma}_{ij}$, $\bar{\Gamma}^i_{jk}$, $\bar{D}_i$, $\bar{R}_{ij}$, etc.

Chapter 2

EMDA Chapter 1

2.1 Introduction

The increasing sensitivity of gravitational-wave detectors has enabled progressively more stringent tests of General Relativity (GR) [20–22]. Recent advances in gravitational-wave astronomy have led to the detection of signals from compact binary mergers, granting direct access to strongly gravitating, nonlinear regimes. All observed mergers to date are consistent with the predictions of GR [4, 15, 23].

Despite this agreement, the long-standing tension between GR and quantum mechanics remains unresolved. The broader field of quantum gravity seeks to address this discrepancy, motivating the exploration of alternative theories of gravity. Examples include Horndeski gravity [24], Lovelock gravity [25, 26], scalar Gauss–Bonnet gravity [27–29], and Einstein–Maxwell–Dilaton–Axion (EMDA) theory [30, 31].

In this work, we focus on the EMDA framework, examining the stability of its black-hole solutions as a prerequisite for producing inspiral waveforms.

As detector sensitivity improves and additional signals are recorded, increasingly precise tests of GR will become possible [32, 33]. Waveforms predicted by alternative theories play a dual role: they guide searches for new signatures and provide benchmarks for testing those theories. Such comparisons may ultimately reveal measurable deviations from GR. Alternative theories generally lack the extensive waveform catalogs available in GR; current efforts often rely on phenomenological approaches, including post-Newtonian expansions and numerical-relativity simulations [34, 35].

In standard GR, black holes are “bald”—that is, they are completely described by their mass, spin, and charge [36, 37]. By contrast, several extensions of GR predict scalarized black holes. For instance, recent studies have examined scalarization during mergers in scalar Gauss–Bonnet gravity [28, 29]. The EMDA theory, which arises in the low-energy limit of heterotic string theory, introduces both a scalar dilaton and an axion while retaining a well-posed initial-data formulation.

The existence of scalar hair in EMDA depends on charge and spin. In the limit of vanishing axion coupling, EMDA reduces to the simpler Einstein–Maxwell–Dilaton (EMD) theory.

Although numerical simulations of binary black holes in the EMDA theory remain limited, progress has been made in the Einstein–Maxwell–Dilaton setting [38, 39], establishing a foundation for extending such studies to the full EMDA framework. It is not immediately clear whether the scalar hair predicted by the EMDA theory remains stable or eventually radiates away, leaving behind a remnant black hole characterized solely by its mass, spin, and charge. Previous studies have explored the related phenomena of scalarization and descalarization in other modified gravity theories [40–42]. EMDA is a particularly compelling theory as it allows for direct tests of low energy string theory. Establishing the stability of Kerr–Sen black holes therefore represents an important direction for future investigation.

Previous work has explored astrophysical observations for potential evidence of Kerr–Sen black holes. Analyses of Event Horizon Telescope (EHT) shadow data suggest that Sgr A* favors the Kerr–Sen solution, while M87* favors Kerr but remains consistent with Kerr–Sen within current observational uncertainties [43, 44].

In this study, we extend these investigations by simulating head-on mergers governed by the full EMDA equations. We analyze the resulting waveforms and examine the stability of the remnants, thereby gaining insight into the persistence of scalar hair within the theory. Specifically, we evolve several head-on collisions of black holes in EMDA to determine the conditions under which the dilaton and axion fields survive the merger. An additional key question is to determine under what conditions initially unscalarized black holes will undergo scalarization.

2.2 Einstein–Maxwell–Dilaton–Axion

EMDA arises from low-energy string theory. This includes a U(1) gauge field and a scalar field that couples to the gauge field. In particular the action for low-energy heterotic string theory is::

$$S = \int d^4x \sqrt{-g} \left[\frac{R}{8\pi G} - 2(\nabla\phi)^2 - e^{-2\alpha_0\phi} F^2 - \frac{1}{2} e^{4\alpha_1\phi} (\nabla\kappa)^2 - \kappa F_{ab} (*F)^{ab} \right]. \quad (2.1)$$

Here, g denotes the determinant of the metric, R the Ricci scalar, ϕ the dilaton, F_{ab} the Faraday tensor, and κ the axion. In the absence of charge, the equations reduce to the standard Einstein equations. In contrast to the no-hair theorem, the EMDA theory admits two additional fields—the dilaton ϕ and the axion κ —which couple to the electromagnetic field and together constitute the “hair.” The parameters α_0 and α_1 set the coupling strengths; in this work we take $\alpha_0 = \alpha_1 = 1$, corresponding to the low-energy limit of heterotic string theory. Setting $\alpha_0 = \alpha_1 = 0$ reduces the theory to Einstein–Maxwell with minimally coupled scalar and axion fields.

To perform numerical evolutions of Kerr–Sen black holes, we formulate the equations as a Cauchy problem and use the BSSN formulation [45–47] of the Einstein equations, supplemented with the necessary terms to account for electromagnetic fields, axion and dilaton. We also employ the puncture gauge condition [48, 49]. Initial data are prepared for head-on black hole collisions. The full equations of motion and the 3 + 1 decomposition are presented in Appendix H.1.

2.3 Numerical methods

2.3.1 Code Description

For these simulations, we modified the existing numerical-relativity framework Dendro-GR. Dendro-GR is a portable code that combines an octree-based adaptive multiresolution framework with automatic code generation for solving the EMDA equations. Spatial derivatives are

computed using sixth-order finite-difference stencils, and time integration is performed with a fourth-order Runge–Kutta scheme. More details on Dendro–GR can be found in [12, 50–52].

2.3.2 Initial Data

Valid initial data must satisfy the constraint equations; in particular, we must satisfy the Hamiltonian, momentum, and Maxwell constraints. In natural units, these are given by

$$R + K^2 - K_{ij}K^{ij} = 16\pi\rho, \quad (2.2)$$

$$D_j(K^{ij} - \gamma^{ij}K) = 8\pi J^i, \quad (2.3)$$

where $\mathcal{R}$ is the three-dimensional Ricci scalar associated with the metric, K_{ij} is the extrinsic curvature, K is the trace of the extrinsic curvature, ρ is the energy density, and J^i is the current density.

$$D_i E^i = 0, \quad D_i B^i = 0, \quad (2.4)$$

The energy and current densities are defined as

$$\begin{aligned} \rho \equiv & \Pi^2 + (D\phi)^2 + e^{-2\alpha_0\phi} (E_a E^a + B_a B^a) \\ & + \frac{1}{4} e^{4\alpha_1\phi} (\Xi^2 + (D\kappa)^2), \end{aligned} \quad (2.5)$$

$$J^i \equiv 2\Pi D^i \phi + 2e^{-2\alpha_0\phi} \epsilon^{ijk} E_j B_k + \frac{1}{2} e^{4\alpha_1\phi} \Xi D^i \kappa. \quad (2.6)$$

The EMDA theory possesses a mathematical structure that admits well-posed initial data. The formulation of such data has not been extensively investigated in the literature, although there have been some approaches to generating initial data for charged black holes [53–57]. In this work, we adapt techniques originally developed for constructing the initial data of charged black holes. Specifically, we employ an elliptic solver based on a modified version of the two-charged-punctures solver introduced in [58], augmented to incorporate contributions from the dilaton. A more detailed discussion of the initial-data construction is provided in Appendix H.2.

To analyze the numerical data, we compute the Newman–Penrose scalar Ψ_4 , as well as the radiated electromagnetic quantities, extracting their multipole moments at $r \approx 100M$. Further details on the Waveform extraction are in Appendix C.

2.3.3 Simulation Setup

A summary of the black hole parameters used in the simulations is provided in Appendix H.4.

2.4 Results

We performed tests on a number of black holes in the nonextremal limit of the parameter space; our search was limited to equal–mass–ratio black holes with low to moderate charge and low spin. We present here a representative example of the waveform and energy radiated. The total mass of the system is 1, and the charge of the system lies between $0.0625 \leq \frac{Q}{M} \leq 0.5$, with spins in the range $0.05 \leq \chi \leq 0.2$. A full summary can be seen in Appendix C. The waveforms of a head-on collision corresponding to two identical black holes with $\chi = 0.2$ and $\frac{Q}{M} = 0.5$ at an initial separation of $x = 8$ are shown in Fig. ???. The results from this configuration are largely representative of the overall behavior observed in our tests.

The radiated energy as a function of time is shown in Fig. 2.1.

In this case, the gravitational wave is the dominant form of energy released. The electromagnetic radiation is the next most significant, followed by the dilaton radiation. The contribution from the axion remains comparatively small. We see that the energy from the scalar fields quickly dies off after the merger occurs. This indicates that once the merger has taken place, the field remains in a relatively stable state. In our experiments, we find that the scalar fields persist and do not radiate away after the merger.

A non-trivial axion field persists after the merger, as illustrated in Fig. ??.

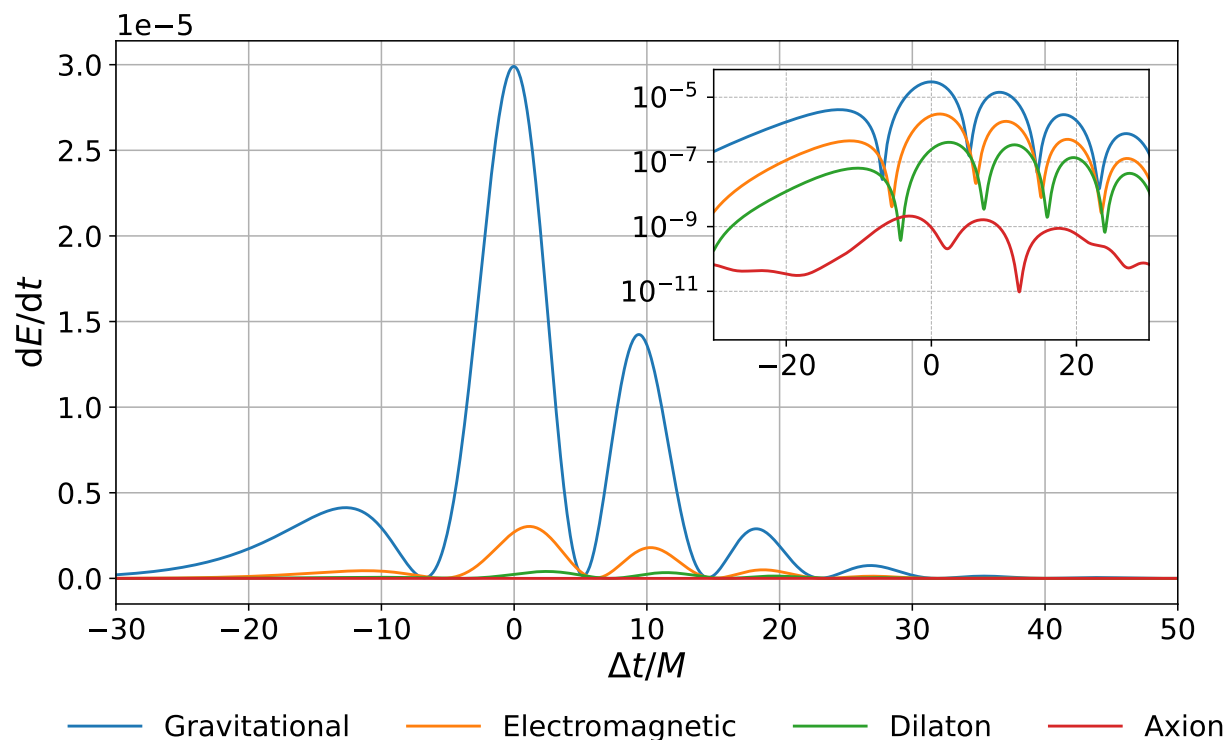


Figure 2.1 Radiated fluxes for the case $\chi = 0.2$ and $Q/M = 0.5$. The curves have been aligned such that $t = 0$ corresponds to the global maximum. The inset plots show the same data with the y-axis on a logarithmic scale.

2.4.1 Initially Unscalarized Black Holes

To further assess the stability of scalar hair, we conducted a series of tests initialized with a Gaussian distribution of axion and dilaton fields centered at the origin, with amplitudes ranging from 10^{-1} to 10^{-7} . These initial perturbations propagate outward, enabling us to examine whether initially unscalarized black holes can acquire and sustain scalar fields. Our results confirm that they do: a Kerr–Newman black hole in the EMDA theory rapidly scalarizes, developing both dilaton and axion hair.

Remarkably, even in the absence of any initial scalar fields in the background, the system evolves to generate nontrivial scalar structure. This demonstrates that scalarization occurs dynamically and does not require preexisting scalar hair. The conserved quantities, including the total mass and

charge, remain effectively constant throughout the evolution. Variations in the conserved quantities over the course of the evolution in both mass and charge are below 0.2%. This indicates that our quantities remain stable and conserved overtime. Figure ?? illustrates the evolution of the scalar fields for the case without initial scalar hair.

2.5 Conclusion

We have simulated a range of different parameters for the head-on collisions of Kerr–Sen black holes. We have primarily been concerned with the persistence of hair. In summary, our simulations show that the scalar fields do not radiate away entirely, so long as there is a nonzero charge (and spin for the axion). We have found that Kerr–Sen black holes are stable. In particular, we find that if the remnant black hole has a nonzero charge, the dilaton field persists. If it has both nonzero charge and spin, the axion field also persists after merger. These results suggest that Kerr–Sen black holes are stable under the conditions and time frames we considered.

A natural follow-up study would be the analysis of waveforms produced by inspiraling black holes, an area where the available results remain relatively limited. In this paper, we have further assumed that we are working within the low-energy sector of string theory. A more detailed investigation of the theory space parameterized by α_0 could be intriguing; in particular, the coupling $\alpha_0 = \sqrt{3}$ corresponds to the Kaluza–Klein theory [59]. Future work may explore interpolations between these theories.

Chapter 3

EMDA Chapter 2

Chapter 4

EMDA Chapter 3

Chapter 5

CFD Chapter 1

5.1 CFD

Chapter 6

Conclusion

6.1 Conclusion

Appendix A

Mathematical Overview

Appendix B

EMDA Black Holes

Appendix C

Computation of Ψ_4

C.1 Computation of Ψ_4

The waveforms we plot are often given in terms of what is known as the Newman–Penrose scalar this is one way to decompose the gravitational waves that come off of our system. In this appendix, we introduce the Newman–Penrose scalar Ψ_4 and describe how it is computed in numerical relativity.

We begin by defining a tetrad. We take the timelike vector to be normal to the spacelike hypersurface. In the asymptotic region, we may construct a set of spatial vectors resembling spherical coordinates and then orthonormalize them using the Gram–Schmidt procedure.

$$\begin{aligned} u^a &= (0, x, y, z), \\ v^a &= (0, xz, yz, -x^2 - y^2), \\ w^a &= (0, -y, x, 0). \end{aligned} \tag{C.1}$$

Using the spatial metric h_{ij} , we orthonormalize them with respect to it: In particular we define new orthonormal space like vectors as:

$$\begin{aligned} {}^{(1)}e^a &= \frac{u^i}{\|u\|}, \\ {}^{(2)}e^a &= \frac{v^i - \langle {}^{(1)}e, v \rangle {}^{(1)}e^i}{\|v - \langle {}^{(1)}e, v \rangle {}^{(1)}e\|}, \\ {}^{(3)}e^a &= \frac{w^i - \langle {}^{(1)}e, w \rangle {}^{(1)}e^i - \langle {}^{(2)}e, w \rangle {}^{(2)}e^i}{\|w - \langle {}^{(1)}e, w \rangle {}^{(1)}e - \langle {}^{(2)}e, w \rangle {}^{(2)}e\|}. \end{aligned} \tag{C.2}$$

where we define the inner product and the norm as

$$\langle u, v \rangle \equiv h_{ij} u^i v^j \tag{C.3}$$

$$\|u\| \equiv \sqrt{\langle u, u \rangle} \tag{C.4}$$

We identify ${}^{(1)}e^a = r^a$, ${}^{(2)}e^a = \theta^a$, and ${}^{(3)}e^a = \phi^a$. Using these, we construct the null tetrad:

$$\begin{aligned}
\ell^a &= \frac{1}{\sqrt{2}}(n^a + r^a), \\
\tilde{n}^a &= \frac{1}{\sqrt{2}}(n^a - r^a), \\
m^a &= \frac{1}{\sqrt{2}}(\theta^a + i\phi^a), \\
\bar{m}^a &= \frac{1}{\sqrt{2}}(\theta^a - i\phi^a).
\end{aligned} \tag{C.5}$$

C.2 Null Tetrads and Their Properties

A null tetrad is a set of four vectors

$$\{\ell^a, n^a, m^a, \bar{m}^a\}$$

forming a basis for the tangent space at each point in spacetime. Here ℓ^a and n^a are real null vectors, while m^a is a complex null vector and $\bar{m}^a$ is its complex conjugate. This formalism is central to the Newman–Penrose approach to general relativity [2, 18].

C.2.1 Null Conditions

Each vector in the tetrad is null:

$$\ell^a \ell_a = 0, \quad n^a n_a = 0, \quad m^a m_a = 0, \quad \bar{m}^a \bar{m}_a = 0. \tag{C.6}$$

C.2.2 Orthogonality Relations

The tetrad vectors satisfy the following orthogonality conditions:

$$\ell^a m_a = \ell^a \bar{m}_a = n^a m_a = n^a \bar{m}_a = 0. \tag{C.7}$$

C.2.3 Normalization

The only non-vanishing inner products are

$$\ell^a n_a = -1, \quad m^a \bar{m}_a = 1, \quad (C.8)$$

where we have assumed the metric signature $(-, +, +, +)$.

C.3 The Weyl Tensor

We shall mostly be concerned with tensors in four dimensions or less. The algebraic identities lead to the following special cases for the curvature tensor:

1. If $n = 1$, $R_{abcd} = 0$.
2. If $n = 2$, R_{abcd} has one independent component — essentially R .
3. If $n = 3$, R_{abcd} has six independent components — essentially R_{ab} .
4. If $n = 4$, R_{abcd} has twenty independent components — ten of which are given by R_{ab} and the remaining ten by the Weyl tensor.

The **Weyl tensor**, or **conformal tensor**, C_{abcd} is defined in n dimensions ($n \geq 4$) by

$$\begin{aligned} C_{abcd} = R_{abcd} &+ \frac{1}{n-2} (g_{ad}R_{cb} + g_{bc}R_{da} - g_{ac}R_{db} - g_{bd}R_{ca}) \\ &+ \frac{1}{(n-1)(n-2)} (g_{ac}g_{db} - g_{ad}g_{cb}) R. \end{aligned} \quad (C.9)$$

Thus, in four dimensions, this becomes

$$\begin{aligned} C_{abcd} = R_{abcd} &+ \frac{1}{2} (g_{ad}R_{cb} + g_{bc}R_{da} - g_{ac}R_{db} - g_{bd}R_{ca}) \\ &+ \frac{1}{6} (g_{ac}g_{db} - g_{ad}g_{cb}) R. \end{aligned} \quad (C.10)$$

The Weyl tensor possesses the same symmetries as the Riemann tensor, namely,

$$C_{abcd} = -C_{abdc} = -C_{bacd} = C_{cdab}, \quad (\text{C.11})$$

and

$$C_{abcd} + C_{adbc} + C_{acdb} \equiv 0. \quad (\text{C.12})$$

The Newman–Penrose scalar Ψ_4 is defined as

$$\Psi_4 = C_{abcd} n^a \bar{m}^b n^c \bar{m}^d. \quad (\text{C.13})$$

All parts vanish except for the Reimann tensor term. we will now decompose the this and look at several parts in turn. starting with:

$$\Psi_4 = R_{abcd} n^a \bar{m}^b n^c \bar{m}^d. \quad (\text{C.14})$$

Expanding the projection onto the hypersurface, we obtain

$$\Psi_4 = \perp (R_{abcd} n^a n^c) \bar{m}^b \bar{m}^d - \perp (R_{abcd} n^a) r^c \bar{m}^b \bar{m}^d. \quad (\text{C.15})$$

Using the Gauss, Codazzi, and Ricci relations, we have

$$\perp R_{abcd} = {}^{(3)}R_{abcd} + K_{ac} K_{bd} - K_{ad} K_{bc}, \quad (\text{C.16})$$

$$\perp (R_{abcd} n^a) = D_c K_{bd} - D_d K_{bc}, \quad (\text{C.17})$$

$$\perp (R_{abcd} n^a n^c) = \mathcal{L}_n K_{bd} + K_{bc} K^c_d + \frac{1}{\alpha} D_b D_d \alpha. \quad (\text{C.18})$$

Substituting these relations, we find

$$\begin{aligned}\Psi_4 = & \left({}^{(3)}R_{bd} + K K_{bd} - K_{bc} K^c_d \right) \bar{m}^b \bar{m}^d \\ & + (D_c K_{bd} - D_d K_{bc}) \bar{m}^b \bar{m}^d r^c.\end{aligned}\tag{C.19}$$

Expressing these quantities in terms of the BSSN variables, we use

$$K_{ab} = e^{4\phi} \left(\tilde{A}_{ab} + \frac{1}{3} \tilde{\gamma}_{ab} K \right).\tag{C.20}$$

The full expressions in terms of BSSN variables are lengthy; here we present the final result.

The real part is

$$\begin{aligned}\mathcal{R}(\Psi_4) = & {}^{(3)}R_{ab} M^{ab} - 4\pi G S_{ab} M^{ab} \\ & + \left(\chi^{-1} \tilde{A}_{ab} r^a r^b + \frac{1}{3} K \right) \chi^{-1} \tilde{A}_{cd} M^{cd} \\ & - \chi^{-2} \left[(\tilde{A}_{ab} r^a m^b)^2 - (\tilde{A}_{ab} r^a \bar{m}^b)^2 \right] + \dots\end{aligned}\tag{C.21}$$

and the imaginary part is

$$\mathcal{I}(\Psi_4) = - \left[{}^{(3)}R_{ab} N^{ab} - 4\pi G S_{ab} N^{ab} + \dots \right].\tag{C.22}$$

Ψ_4 is not what we directly measure at ligo we instead measure something related called the strain. We can calculate the strain from Ψ_4

We calculate the strain as follows

****include bit about strain****

****include mismatch notes****

Appendix D

Dendro Notes

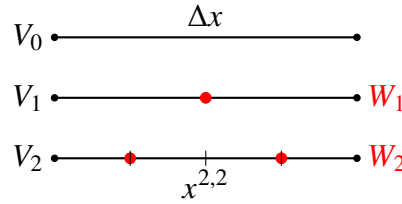
D.1 Wavelet adaptive mesh refinement

We can construct a sparse representation of the fields using a set of nested dyadic grids:

$$V_j = \{x^{j,k} = 2^{-j} k \Delta x\}, \quad (\text{D.1})$$

where V_j denotes the grid level, and $x^{j,k}$ denotes the points on the grids.

I show an example below: the red dots indicate the points included only in the alternative grids W_j . Notice that all points in a given V_j will also appear in all subsequent refinement levels. The grids do not need to separately save the points already included at coarser levels; the points unique to finer grids are highlighted in red.



We start with a field defined on the points of the grid. The field is represented by $u^{j,k}$. We can extend the values to higher levels by interpolation. For points already included at coarser resolutions, we have $u^{j+1,2k} = u^{j,k}$. For new points, interpolation gives

$$u^{j+1,2k+1} = \sum_m h_{j,m}^{j+1,2k+1} u^{j,m}. \quad (\text{D.2})$$

We can represent this interpolation with a series of basis functions:

$$\phi_{j,k}(x) = \sum_l c^l \phi_{j+1,l}(x), \quad (\text{D.3})$$

where c^l are the interpolation weights.

To represent the full grid, including the alternative grids W_j (the red dots above), we can write:

$$u(x) = \sum_{k \in S_0} u^{0,k} \phi_{0,k}(x) + \sum_{j=1} \sum_{k \in S_j} d^{j,k} \phi_{j,k}(x), \quad (\text{D.4})$$

where $d^{j,k}$ are the wavelet coefficients. The $u^{0,k}$ values are the field values at base level points, $u^{0,k} = u(x^{0,k})$, and $u^{j,k} = u(x^{j,k})$.

The wavelet coefficients are computed by comparing the interpolation from the previous level to the actual field value:

$$d^{j,k} = u^{j,k} - P(x^{j,k}, j-1), \quad (\text{D.5})$$

where $P(x^{j,k}, j-1)$ is the interpolated value from level $j-1$.

The wavelet coefficient measures how well the interpolation captures the field at that point. We can set a threshold ϵ such that refinement occurs only where

$$|d^{j,k}| > \epsilon.$$

D.2 Spacefilling curves and the octree

Appendix E

CFD Notes

Appendix F

Software Notes

Appendix G

EMDA Code Setup on MaryLou

G.1 Building `emda-gr`

To build `emda-gr`, the following external packages are required:

- **C/C++11 or higher** (tested with GNU and Intel compilers)
- **MPI implementation** (tested with MVAPICH2, OpenMPI, and Intel MPI)
- **BLAS/LAPACK** (tested with OpenBLAS and Intel MKL)
- **GSL** (GNU Scientific Library); set `GSL_ROOT_DIR` in CMake to enable automatic detection

The modules currently loaded on MaryLou are:

```
andrewc0@login03:$ ml
```

Currently Loaded Modules:

- 1) gcc/14.1
- 2) python/3.12
- 3) cmake/3.28

- 4) gcc-runtime/13.2.0-g4hc7ev
- 5) eigen/3.4.0-6kqvaiac
- 6) openmpi5-GR
- 7) gsl-GR
- 8) mkl-GR
- 9) rclone/1.65.2-tyg4lvy

To ensure these dependencies are always available, it is convenient to load them automatically in your shell configuration. Open your `~/.bash_profile`:

```
vim ~/.bash_profile
```

An example `.bash_profile` is shown below. Additional customizations may be added as needed.

```
# --- Modules ---
module purge
module use /grphome/grp_GR/.modulefiles
module load gcc/14.1 python/3.12 cmake/3.28 eigen/3.4.0-6kqvaiac
module load openmpi5-GR gsl-GR mkl-GR rclone

# --- GSL ---
export GSL_DIR=/grphome/grp_GR/apps/gsl-2.8
export GSL_ROOT_DIR=/grphome/grp_GR/apps/gsl-2.8
export LD_LIBRARY_PATH="${GSL_DIR}/lib:${LD_LIBRARY_PATH:-}"

# --- MKL ---
if [ -n "${MKLROOT:-}" ] && [ -d "${MKLROOT}/lib/intel64" ]; then
```



```
export LD_LIBRARY_PATH="${MKLRROOT}/lib/intel64:${LD_LIBRARY_PATH:-}"
fi

# --- GCC 14 runtime (libstdc++, libgcc_s) ---
# Required to avoid GLIBCXX / CXXABI runtime errors
if command -v g++ >/dev/null 2>&1; then
    GCC_LIBDIR="$(dirname "$(g++ -print-file-name=libstdc++.so.6)")"
    GCC_GCCLIBDIR="$(dirname "$(g++ -print-file-name=libgcc_s.so.1)")"

    if [ -d "$GCC_LIBDIR" ]; then
        export LD_LIBRARY_PATH="$GCC_LIBDIR:${LD_LIBRARY_PATH:-}"
    fi

    if [ -d "$GCC_GCCLIBDIR" ]; then
        export LD_LIBRARY_PATH="$GCC_GCCLIBDIR:${LD_LIBRARY_PATH:-}"
    fi
fi

# --- History settings ---
HISTSIZE=1000000
HISTFILESIZE=2000000
HISTCONTROL=ignoredups:erasedups
HISTIGNORE="ls:bg:fg:history"
shopt -s histappend

# Save history after every command
```

```
PROMPT_COMMAND="history -a; history -c; history -r; $PROMPT_COMMAND"
```

After reloading your shell, verify that the modules have been loaded correctly:

```
m1
```

The output should match the module list shown above.

G.2 Building the Solver

First, clone the EMDA repository. Create a project directory if desired, then run:

```
git clone git@github.com:dfvankomen/emda-gr.git
cd emda-gr
```

CMake will now automatically fetch `dendrolib`.

Create a build directory:

```
mkdir build
cd build
```

You may then configure the project using:

```
ccmake ..
```

CMake will locate the required libraries and present a configuration interface. After configuration, generate the build files. At present, the only settings you may wish to toggle is the choice between the EMDA-EMD-EMGR systems of equations. The second is the CPU architecture we will be running on. For the purposes of the first run, let us use EMDA.

We will also need to specify the CPU architecture for the target system. For example on MaryLou, we normally use the `physics2` or `m12` partitions, both of which consist of 64-core AMD EPYC

7763 (2.45 GHz) CPUs. Therefore, we should build for the Zen 3 architecture by setting `CPU_ARCH` to `znver3`.

Once the build files are generated, compile the code using:

```
make -j
```

G.2.1 Build Outputs

Once the build completes, you should have both `tpid` and `emdaSolver` binaries located in one of the following directories:

```
emda-gr/build/emdaSolver/
```

At this point, the solver is ready, and we can proceed to running simulations.

G.3 Runs and Job Scripts

With the solver built, we now need to move from your home directory into the scratch directory. The scratch directory varies by supercomputer. Consult the Supercomputer documentation I will show how this is done on Marylou.

First, change into your scratch area:

```
cd /nobackup/autodelete/usr/YOURUSERNAME/
```

Create a directory for your runs (name it however you like), then enter it:

```
mkdir EMDA_runs
```

```
cd EMDA_runs
```

Next, copy over the binaries we just built, along with a parameter (TOML) file. (The ~ in these commands refers to your home directory.)

Copy emdaSolver:

```
cp ~/path/to/emda-gr/build/build_noavx/emdaSolver/emdaSolver ./
```

Copy tpid:

```
cp ~/path/to/emda-gr/build/build_noavx/emdaSolver/tpid ./
```

Copy the parameter file:

```
cp ~/path/to/emda-gr/emda_simplified.param.toml ./
```

Verify that everything copied correctly:

```
ls
```

You should see:

```
emda_simplified.param.toml  emdaSolver  tpid
```

Now create output directories for the run:

```
mkdir cp prof vtu bah
```

The vtu is where visual data that we can see in paraview is saved, cp is where checkpoints are saved prof is where the gravitational and other waves are saved as well as the ADM Mass charge and the constraint solutions are saved. (We will add another section later describing the relevant parameter-file settings.)

G.3.1 Submitting a Run with Slurm

To run on the cluster, create a job script. Use vim:

```
vim Job.sb
```

Paste the following:

```
#!/bin/zsh

#SBATCH --time=72:00:00

#SBATCH --ntasks=256

#SBATCH --nodes=2

#SBATCH --mem-per-cpu=4096M

#SBATCH -J EMDA_RUN

#SBATCH --mail-user=youremail@email.com

#SBATCH --mail-type=END,FAIL

#SBATCH --partition=m12


echo "Starting run"


source ~/.bash_profile

./tpid emda_simplified.param.toml 256 &> TPID.txt

mpirun -np 256 ./emdaSolver emda_simplified.param.toml &>> Inspiral.txt


echo "Run completed"
```

Note that the semicolon means “do this before moving on.” We need this behavior so that the TPID solver finishes generating the initial data before proceeding. TPID will produce a binary file called `rit_q2_tpid_sol.bin`, which `emdaSolver` will then read and evolve.

On the m12 nodes, each node has 128 CPU cores, so you should scale the number of nodes accordingly (e.g., 256 tasks typically corresponds to 2 nodes).

You can submit the job using:

```
sbatch Job.sb
```

Once submitted, the job will enter the queue and will begin running when resources become available. You can check the status of your jobs with:

```
squeue | grep yourusername
```

After the job starts running, you can monitor the output by inspecting the text files generated during execution, for example:

```
cat TPID.txt
```

And that's it—the simulation is running.

The only remaining step is to explain the parameter file:

G.4 Par-file Explanation

The parameter file is read by both the `tpid` solver and `emdaSolver`. It allows us to change solver settings without having to rebuild the code each time, which is extremely useful.

Below is an overview of the most important settings. If a parameter is not mentioned here, it is probably safe to leave it at its default value.

```
# Whether or not the solver should attempt to restore from a checkpoint  
EMDA_RESTORE_SOLVER = false
```

This determines whether the solver will attempt to pick up where it left off by restoring from a checkpoint.

```
EMDA_IO_OUTPUT_FREQ = 400
```

This controls how often VTU output is written. Smaller values produce output more frequently but increase storage usage. Adjust this setting to suit your needs.

```
EMDA_REMESH_TEST_FREQ = 200
```

This sets how often the solver invokes the AMR algorithm to determine whether the grid resolution needs to change. Setting this too often can add noise on the grid. updating too infrequently can mean the dynamics are not well captured by the grid. Around 200 seem to work well.

```
EMDA_GW_EXTRACT_FREQ = 25
```

This controls how often gravitational waves are extracted, along with black hole positions and masses. This step is not particularly expensive, either computationally or in terms of storage, so values in the range 50–100 are typically reasonable.

```
EMDA_CHECKPT_FREQ = 100
```

This specifies the number of time steps between checkpoints. This is pretty cheap computationally so this can be small.

```
# VTU file prefix
```

```
EMDA_VTU_FILE_PREFIX = "vtu/emda_gr"
```

```
# Checkpoint file prefix
```

```
EMDA_CHKPT_FILE_PREFIX = "cp/emda_cp"
```

```
# Profiling file prefix
```

```
EMDA_PROFILE_FILE_PREFIX = "prof/emda_prof"
```

These options control where different types of output data are stored.

```
# Which evolution variables to output
EMDA_VTU_OUTPUT_EVOL_INDICES = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14,
15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34,
35]
```

```
# Number of evolution variables to output
EMDA_NUM_EVOL_VARS_VTU_OUTPUT = 36
```

These settings determine which evolution variables (e.g., metric components, extrinsic curvature, etc.) are written to the VTU files. If you need to reduce storage usage, you may omit some of these variables.

```
# Which constraint variables to output
EMDA_VTU_OUTPUT_CONST_INDICES = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13]
```

```
# Number of constraint variables to output
EMDA_NUM_CONST_VARS_VTU_OUTPUT = 14
```

These are similar to the evolution-variable settings, but apply to constraint quantities such as the Hamiltonian and momentum constraints.

```
# Whether to isolate an X slice
EMDA_VTU_X_SLICE = false
# Whether to isolate a Y slice
EMDA_VTU_Y_SLICE = false
# Whether to isolate a Z slice
EMDA_VTU_Z_SLICE = true
# Any combination may be set to true
```


In general, writing VTU output for the entire 3-D grid is expensive, so we typically output a 2-D slice instead. You should enable only one of these slice options at a time; enabling multiple slices will produce unintuitive cuts through the grid. For most purposes, leaving the Z slice enabled is a good choice.

```
EMDA_DENDRO_GRAIN_SZ = 50
```

This controls how many grid blocks are assigned to each processor. Some analysis has been done to estimate optimal values, but in practice this parameter should generally be kept between 25 and 200.

```
EMDA_DENDRO_AMR_FAC = 0.2
```

This controls how sensitive the AMR algorithm is to refining or coarsening the grid.

```
EMDA_ELE_ORDER = 6
```

Loosely speaking, this determines how much information each block shares with its neighbors. Higher values increase communication and can slow the code down. For EMDA, this should currently be left at 6.

```
# Minimum depth of the mesh/octree
```

```
EMDA_MINDEPTH = 4
```

```
# Maximum depth of the mesh/octree
```

```
EMDA_MAXDEPTH = 15
```

These are very important parameters. They determine how coarse (EMDA_MINDEPTH) and how fine (EMDA_MAXDEPTH) the grid is allowed to become. Note that the CFL factor is set by the finest level of the grid, so increasing EMDA_MAXDEPTH significantly increases the runtime. Each increase

in EMDA_MAXDEPTH halves the grid spacing at the finest level. Each black hole needs a minimum of 50 or more points accross the horizon to be adequately resolved. Maxdepth 15 for a spaceteim with a total mass of 1, with an equal mass ratio merger on a ± 400 grid is adequately resolved. as q goes up this increases for a q=2 you need a maxdepth of 16.

```
#####
```

```
# REFINEMENT STRATEGY
```

```
# Which refinement mode to use
```

```
# 0 - WAMR
```

```
# 1 - EH
```

```
# 2 - EH_WAMR
```

```
# 3 - BH_LOC
```

```
# 4 - BH_WAMR
```

```
EMDA_REFINEMENT_MODE = 4
```

This parameter controls how the grid is refined. it is recommended to leave this set to 4 for now.

```
EMDA_AMR_R_RATIO = 1.618033988749895
```

This determines how quickly refinement around a black hole falls off with radius. The default value is reasonable and should be left unchanged initially.

```
# Wavelet Tolerance Function to use (see grUtils.cpp for adding more)
```

```
# 1,2 - deprecated
```

```
# 3 - center highly refined; low refinement elsewhere eases over time
```

```
# 4 - start with refinement centered at BHs, then widen to r=0 focused
```

```
# 5 - focus refinement in self-similar ways about each BH
```

```
# 6 - refine well in  $r \leq t$  and coarsely in  $r > t$ 
```

```
EMDA_USE_WAVELET_TOL_FUNCTION = 6
```

This controls how the refinement tolerance varies across the grid. Function 6, which was developed by William, works well in practice and is recommended. More details on this can be found [52]

```
# Minimum wavelet tolerance
```

```
EMDA_WAVELET_TOL = 0.00001
```

```
# Gravitational wave refinement tolerance
```

```
EMDA_GW_REFINE_WTOL = 0.0001
```

```
# Maximum wavelet tolerance
```

```
EMDA_WAVELET_TOL_MAX = 0.001
```

```
# Wavelet tolerance radii (for shell-based tolerance)
```

```
EMDA_WAVELET_TOL_FUNCTION_R0 = 3.0
```

```
EMDA_WAVELET_TOL_FUNCTION_R1 = 12.0
```

The minimum wavelet tolerance is the most important of these parameters, as it sets the strictest refinement criterion over the entire grid. The remaining parameters depend on the chosen tolerance function; the defaults are reasonable, but may be adjusted if needed.

```
# Indices of variables used for refinement (see grDef.h for definitions)
```

```
EMDA_REFINE_VARIABLE_INDICES = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14,  
15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34,  
35]
```

```
# Number of variables used for refinement
```

```
EMDA_NUM_REFINE_VARS = 36
```

we can choose not to test every evolution variable when deciding whether to refine the grid. These parameters control which variables are used to trigger refinement.

```
# CFL (Courant--Friedrichs--Lewy) factor
```

```
EMDA_CFL_FACTOR = 0.25
```

```
# Simulation start time
```

```
EMDA_RK_TIME_BEGIN = 0
```

```
# Simulation end time
```

```
EMDA_RK_TIME_END = 1000
```

These parameters control the CFL factor and the total coordinate time for which the simulation is run. The remaining Runge–Kutta settings are typically fine at their default values.

```
#####
```

```
# EMDA THEORY PARAMETERS
```

```
#####
```

```
EMDA_LAMBDA = [1, 1, 1, 1]
```

```
EMDA_ALPHA_THEORY = [1.0, 1.0]
```

```
EMDA_LF = [1.0, 0.0]
```

```
EMDA_P_EXPO = -1.0
```

```
EMDA_ALPHA_POS_NEG_FLOOR = false
```

```
EMDA_ETA = [0.4, 0.4]
```

```
EMDA_XI = [0, 0, 0]
```

```

CHI_FLOOR = 0.00001
EMDA_TRK0 = 0.0
EMDA_ETA_R0 = 1.31
EMDA_ETA_POWER = [2.0, 2.0]
EMDA_ETADAMP = 1.0

```

```

ETA_CONST = 2.0
ETA_R0 = 30.0
ETA_DAMPING = 1.0
ETA_DAMPING_EXP = 1.0

```

These parameters define the physics of the system, many of which control the gauge choice. The two parameters that we modify are α_0 and α_1 , which select the theory being evolved: $\alpha_0 = \alpha_1 = 1$ corresponds to EMDA, $\alpha_0 = 1, \alpha_1 = 0$ corresponds to EMD, and $\alpha_0 = \alpha_1 = 0$ corresponds to Einstein–Maxwell. The remaining gauge parameters are generally fine at their default values.

```

# Kreiss--Oliger dissipation parameter
KO DISS SIGMA = 0.4

```

This parameter controls the strength of numerical dissipation. Some dissipation is required for stability, and a value of 0.4 is a reasonable starting point.

```

#####
# Select initial data type
# See the bottom of the file for reference

EMDA_ID_TYPE = 0

```

This selects the type of initial data used. A value of 0 uses the TPID solver, though other black hole initial-data options are available.

```
#####
```

```
# Set up EMDA grid extents
```

```
EMDA_GRID_MIN_X = -400.0
```

```
EMDA_GRID_MAX_X = 400.0
```

```
EMDA_GRID_MIN_Y = -400.0
```

```
EMDA_GRID_MAX_Y = 400.0
```

```
EMDA_GRID_MIN_Z = -400.0
```

```
EMDA_GRID_MAX_Z = 400.0
```

These parameters define the physical size of the computational grid. The values shown here are reasonable defaults. We do not want the boundaries to close to where we extract the waves as reflections from the boundaries can contaminate our waveforms.

```
#####
```

```
# Black hole parameters
```

```
EMDA_BH1_AMR_R = 1.0
```

```
EMDA_BH1_CONSTRAINT_R = 2.0
```

```
EMDA_BH1_MAX_LEV = 15
```

```
EMDA_BH1 = {MASS = 0.5, X = 5.0, Y = 0.0, Z = 0.0,
```

```
            V_X = -0.00100282, V_Y = 0.095775311, V_Z = 0.0,
```

```
            SPIN = 0.0, SPIN_THETA = 0.0, SPIN_PHI = 0.0,
```

```
            ELECTRIC_CHARGE = 0.05, MAGNETIC_CHARGE = 0.0}
```

```

EMDA_BH2_AMR_R = 1.0
EMDA_BH2_CONSTRAINT_R = 2.0
EMDA_BH2_MAX_LEV = 15
EMDA_BH2 = {MASS = 0.5, X = -5.0, Y = 0.0, Z = 0.0,
            V_X = 0.00100282, V_Y = -0.095775311, V_Z = 0.0,
            SPIN = 0.0, SPIN_THETA = 0.0, SPIN_PHI = 0.0,
            ELECTRIC_CHARGE = 0.05, MAGNETIC_CHARGE = 0.0}

```

Here we set the parameters for each black hole. `EMDA_BH1_AMR_R` defines the radius within which refinement is forced around each black hole, while `EMDA_BH1_CONSTRAINT_R` specifies the radius excluded when computing Hamiltonian and momentum constraints. `EMDA_BH2_MAX_LEV` sets the maximum refinement level near the black hole.

Note that the `X`, `Y`, and `Z` values do not determine the initial placement of the black holes on the grid. Instead, they specify where the code begins searching for the black hole location and define the excision region. For equal-mass binaries ($q = 1$), these should match $\pm b$ in the TPID parameters.

The electric charge must remain smaller than the mass to avoid exceeding the extremal limit. Magnetic charge should not be used yet, as it is still under development. The spin parameter corresponds to $S = \chi m^2$. TPID reads the charge and spin from this line, but not the mass or position.

Note that `V_X`, `V_Y`, and `V_Z` represent the *momenta*, not the velocities. This is an important distinction.

```
#####
```

```
# GRAVITATIONAL WAVE EXTRACTION PARAMETERS
```

```

EMDA_GW_NUM_RADAI = 6
EMDA_GW_RADAI = [50.0, 60.0, 70.0, 80.0, 90.0, 100.0]

```

```

EMDA_GW_NUM_LMODES = 7
EMDA_GW_L_MODES = [2, 3, 4, 5, 6, 7, 8]

# Radius at which conserved quantities are extracted
EMDA_GW_EXTRACTION_RADIUS = 300.0

# Modes for EM, dilaton, and axion radiation
EMDA_RAD_NUM_LMODES = 6
EMDA_RAD_L_MODES = [1, 2, 3, 4, 5, 6]
#####

```

These parameters control the extraction of gravitational, electromagnetic, dilaton, and axion radiation. We can choose which spherical harmonics we want to extract. `EMDA_GW_NUM_LMODES` applies to gravitational waves, while `EMDA_RAD_NUM_LMODES` applies to all other radiative fields. The mode numbers correspond to the multipole moments being extracted. Although modes can be defined up to $l \approx 10$, in practice the dominant contributions are usually from modes with $l \leq 4$.

Radiation is extracted on spherical surfaces at several radii, which are specified above.

```

# Time steps between apparent-horizon (AEH) solves
AEH_SOLVER_FREQ = 50

```

This parameter controls how often the apparent-horizon solver is invoked. The remaining AEH-related settings should generally be left at their default values.

```

TPID_PAR_B = 5.0
TPID_TARGET_M_PLUS = 0.48594
TPID_TARGET_M_MINUS = 0.48594

```


TPID_PAR_B sets the coordinate separation between the black holes. The target masses are the values used internally by the TPID solver.

TPID_CENTER_OFFSET = {X = 0.0, Y = 0.0, Z = 0.0}

For equal-mass binaries ($q = 1$), this should be left unchanged. For unequal mass ratios, the system should be recentered on the center of mass.

TPID_NPOINTS_A = 40

TPID_NPOINTS_B = 40

TPID_NPOINTS_PHI = 20

These parameters control the resolution of the TPID grid. The values shown here are reasonable defaults.

TPID_GIVE_BARE_MASS = 1

This determines whether the puncture masses are fixed or solved for. Setting this to 0 instructs TPID to solve for the masses. When set to 1, TPID attempts to match the puncture masses to the specified target values.

A recommended procedure is to first estimate the expected masses (e.g., $m = 0.5$ for $q = 1$), run TPID with mass solving enabled, then update the target masses and rerun with TPID_GIVE_BARE_MASS = 1 to finalize the initial data.

TPID_REPLACE_LAPSE_WITH_SQRT_CHI = true

This option should always be enabled when using SSL.

Include dilatonic contributions in rho during TPID solve

TPID_USE_DILATON = true

```
# Interpolated initial data should not be used with TPID_USE_DILATON
# You should also be using the EMD equations in this case
TPID_USE_INTERPOLATED_SOLUTION = false
```

If both options are set to `false`, TPID will generate pure Einstein–Maxwell initial data. Setting `TPID_USE_INTERPOLATED_SOLUTION = true` produces EMD initial data, while `TPID_USE_DILATON = true` produces EMMA initial data.

You should ensure that the α parameters are consistent with the type of initial data being generated, and that CMake is configured to build the corresponding system of equations.

Appendix H

General Notes

H.1 Equations of Motion

We start with the equations of motion for EMDA. We begin with the action in the Einstein frame:

$$S = \int d^4x \sqrt{-g} \left[\frac{R}{8\pi G} - 2(\nabla\phi)^2 - e^{-2\alpha_0\phi} F^2 - \frac{1}{2} e^{4\alpha_1\phi} (\nabla\kappa)^2 - \kappa F_{ab} (*F)^{ab} \right]. \quad (\text{H.1})$$

Here, κ is the axion, related to the three-form potential by $H_{abc} = \frac{1}{2} e^{4\phi} \epsilon_{abc}{}^d \partial_d \kappa$, and α_0 and α_1 parameterize the theory. In this instance, we are interested in the sector of low-energy string theory where $\alpha_0 = \alpha_1 = 1$.

The covariant equations of motion are:

$$0 = \nabla^a \nabla_a \phi + \frac{1}{2} \alpha_0 e^{-2\alpha_0\phi} F^2 - \frac{1}{2} \alpha_1 e^{4\alpha_1\phi} (\nabla\kappa)^2, \quad (\text{H.2})$$

$$0 = \nabla^a \left(e^{4\alpha_1\phi} \nabla_a \kappa \right) - F_{ab} (*F)^{ab}, \quad (\text{H.3})$$

$$0 = \nabla_a \left(e^{-2\alpha_0\phi} F^{ab} \right) + (*F)^{ab} \nabla_a \kappa. \quad (\text{H.4})$$

The Einstein equation takes the form

$$R_{ab} - \frac{1}{2} g_{ab} R = 8\pi G \left[2 \left(\nabla_a \phi \nabla_b \phi - \frac{1}{2} g_{ab} (\nabla\phi)^2 \right) + 2e^{-2\alpha_0\phi} \left(F_{ac} F_b{}^c - \frac{1}{4} g_{ab} F^2 \right) + \frac{1}{2} e^{4\alpha_1\phi} \left(\nabla_a \kappa \nabla_b \kappa - \frac{1}{2} g_{ab} (\nabla\kappa)^2 \right) \right]. \quad (\text{H.5})$$

To the Maxwell constraints we add constraint damping so that violations propagate away at the speed of light. In particular, we introduce two scalar fields Φ and Ψ :

$$\begin{aligned} \nabla^b (F_{ba} + g_{ba} \Psi) &= \kappa_1 n_a \Psi + 2\alpha_0 \nabla^b \phi F_{ba} \\ &\quad + e^{2\alpha_0\phi} (*F)_{ab} \nabla_b \kappa, \end{aligned} \quad (\text{H.6})$$

$$\nabla^b ((*F)_{ba} + g_{ba} \Phi) = \kappa_2 n_a \Phi. \quad (\text{H.7})$$

For the gravitational part we have

$$(\partial_t - \mathcal{L}_\beta) \tilde{\gamma}_{ij} = -2\alpha \tilde{A}_{ij}, \quad (\text{H.8})$$

$$(\partial_t - \mathcal{L}_\beta) \chi = \frac{2}{3}\alpha \chi K, \quad (\text{H.9})$$

$$(\partial_t - \mathcal{L}_\beta) K = [\dots] + 4\pi\alpha (\rho + T), \quad (\text{H.10})$$

$$(\partial_t - \mathcal{L}_\beta) \tilde{A}_{ij} = [\dots] - 8\pi\alpha \left(\chi T_{ij} - \frac{T}{3} \tilde{\gamma}_{ij} \right), \quad (\text{H.11})$$

$$(\partial_t - \mathcal{L}_\beta) \tilde{\Gamma}^i = [\dots] - 16\pi\alpha \chi^{-1} J^i, \quad (\text{H.12})$$

$$\tilde{\Gamma}^i = \tilde{\gamma}^{jk} \tilde{\Gamma}^i_{jk}. \quad (\text{H.13})$$

Here $[\dots]$ denotes the standard right-hand side of the BSSN equations in the absence of source terms.

The quantities T_{ij} , T , J^i , and ρ are defined as

$$\begin{aligned} J^i &\equiv -h^{ia} T_{ab} n^b \\ &= \frac{1}{8\pi} (2\Pi D^i \phi + 2e^{-2\alpha_0 \phi} \epsilon^{ijk} E_j B_k + \frac{1}{2} e^{4\alpha_1 \phi} \Xi D^i \kappa) \end{aligned} \quad (\text{H.14})$$

$$\begin{aligned} T_{ij} &\equiv T_{ab} h^a_i h^b_j \\ &= \frac{1}{8\pi} \left[2 \left[D_i \phi D_j \phi - \frac{1}{2} h_{ij} ((D\phi)^2 - \Pi^2) \right] \right. \\ &\quad \left. - 2e^{-2\alpha_0 \phi} \left[E_i E_j + B_i B_j - \frac{1}{2} h_{ij} (E_k E^k + B_k B^k) \right] \right. \\ &\quad \left. + \frac{1}{2} e^{4\alpha_1 \phi} \left[D_i \kappa D_j \kappa - \frac{1}{2} h_{ij} ((D\kappa)^2 - \Xi^2) \right] \right]. \end{aligned} \quad (\text{H.15})$$

$$T = T^{ij} \gamma_{ij}, \quad (\text{H.16})$$

$$\begin{aligned} \rho &\equiv \frac{1}{8\pi} \left(\Pi^2 + (D\phi)^2 + e^{-2\alpha_0 \phi} (E_a E^a + B_a B^a) \right. \\ &\quad \left. + \frac{1}{4} e^{4\alpha_1 \phi} (\Xi^2 + (D\kappa)^2) \right) \end{aligned} \quad (\text{H.17})$$

For the matter equations we have:

$$(\partial_t - \mathcal{L}_\beta) \phi = -\alpha \Pi. \quad (\text{H.18})$$

$$(\partial_t - \mathcal{L}_\beta)\kappa = -\alpha\Xi. \quad (\text{H.19})$$

$$\begin{aligned} (\partial_t - \mathcal{L}_\beta)\Pi &= \alpha K\Pi \\ &\quad - \alpha\chi \tilde{\gamma}^{ij} \partial_i \partial_j \phi - \chi \tilde{\gamma}^{ij} \partial_i \alpha \partial_j \phi \\ &\quad + \alpha\chi \tilde{\Gamma}^i \partial_i \phi + \frac{\alpha}{2} \tilde{\gamma}^{ij} \partial_i \chi \partial_j \phi \\ &\quad - \frac{\alpha\alpha_0}{\chi} e^{-2\alpha_0\phi} \tilde{\gamma}_{ij} B^i B^j + \frac{\alpha\alpha_0}{\chi} e^{-2\alpha_0\phi} \tilde{\gamma}_{ij} E^i E^j \\ &\quad + \frac{\alpha\alpha_1}{2} e^{4\alpha_1\phi} \chi \tilde{\gamma}^{ij} \partial_i \kappa \partial_j \kappa - \frac{\alpha\alpha_1}{2} e^{4\alpha_1\phi} \Xi^2. \end{aligned} \quad (\text{H.20})$$

$$\begin{aligned} (\partial_t - \mathcal{L}_\beta)\Xi &= \alpha K\Xi \\ &\quad - \alpha\chi \tilde{\gamma}^{ij} \partial_i \partial_j \kappa - \chi \tilde{\gamma}^{ij} \partial_i \alpha \partial_j \kappa \\ &\quad + \alpha\chi \tilde{\Gamma}^i \partial_i \kappa + \frac{\alpha}{2} \tilde{\gamma}^{ij} \partial_i \kappa \partial_j \chi \\ &\quad - 4\alpha\alpha_1 \chi \tilde{\gamma}^{ij} \partial_i \kappa \partial_j \phi + 4\alpha\alpha_1 \Pi\Xi \\ &\quad + \frac{4\alpha}{\chi} e^{-4\alpha_1\phi} \tilde{\gamma}_{ij} B^i E^j. \end{aligned} \quad (\text{H.21})$$

$$\begin{aligned} (\partial_t - \mathcal{L}_\beta)\Psi &= -\alpha \partial_i E^i \\ &\quad + \frac{3\alpha}{2\chi} E^i \partial_i \chi + 2\alpha_0 \alpha E^i \partial_i \phi \\ &\quad + \alpha e^{2\alpha_0\phi} B^i \partial_i \kappa - \alpha\eta_1 \Psi. \end{aligned} \quad (\text{H.22})$$

$$\begin{aligned} (\partial_t - \mathcal{L}_\beta)\Phi &= \alpha \partial_i B^i \\ &\quad - \frac{3\alpha}{2\chi} B^i \partial_i \chi - \alpha\eta_2 \Phi. \end{aligned} \quad (\text{H.23})$$

$$\begin{aligned}
(\partial_t - \mathcal{L}_\beta)E^i &= \alpha K E^i \\
&+ 2\alpha\alpha_0\chi^{1/2}\epsilon^{jik}\tilde{\gamma}_{kl}B^l\partial_j\phi - 2\alpha\alpha_0E^i\Pi \\
&+ \alpha e^{2\alpha_0\phi}\chi^{1/2}\epsilon^{ijk}\tilde{\gamma}_{kl}E^l\partial_j\kappa - \alpha e^{2\alpha_0\phi}\Xi B^i \\
&- \alpha\chi\tilde{\gamma}^{ij}\partial_j\Psi - \chi^{1/2}\epsilon^{jik}\tilde{\gamma}_{kl}B^l\partial_j\alpha \\
&- \alpha\chi^{1/2}\epsilon^{jik}B^l\partial_j\tilde{\gamma}_{kl} - \alpha\chi^{1/2}\epsilon^{jik}\tilde{\gamma}_{kl}\partial_jB^l \\
&+ \alpha\chi^{-1/2}\epsilon^{jik}\tilde{\gamma}_{kl}B^l\partial_j\chi.
\end{aligned} \tag{H.24}$$

$$\begin{aligned}
(\partial_t - \mathcal{L}_\beta)B^i &= \alpha K B^i + \alpha\chi\tilde{\gamma}^{ij}\partial_j\Phi \\
&- \chi^{1/2}\epsilon^{ijk}\tilde{\gamma}_{kl}E^l\partial_j\alpha + \alpha\chi^{-1/2}\epsilon^{ijk}\tilde{\gamma}_{kl}E^l\partial_j\chi \\
&- \alpha\chi^{1/2}\epsilon^{ijk}E^l\partial_j\tilde{\gamma}_{kl} - \alpha\chi^{1/2}\epsilon^{ijk}\tilde{\gamma}_{kl}\partial_jE^l.
\end{aligned} \tag{H.25}$$

We also make use of the slow-start lapse proposed in [60]. Using this approach reduces the amount of computational resources spent refining junk radiation.

H.2 Initial Data

As a first step, we assume conformally flat space; that is, we decompose the metric as

$$\gamma_{ij} = \psi^4 \tilde{\gamma}_{ij}, \tag{H.26}$$

where a tilde denotes conformal variables and ψ is the conformal factor. The assumption of conformal flatness simplifies the following computations and is a reasonable choice for the systems we study. However, Kerr spacetimes are known not to admit conformally flat slices [61, 62]. Alternative approaches have been proposed to construct initial data for highly spinning black holes [63].

Notwithstanding this, Bowen–York black holes can reach spin values of order $\chi \sim 0.9$ [64]. Since our simulations remain below this limit, the approximation should be acceptable.

We nevertheless anticipate the appearance of “junk radiation” at the beginning of the simulation, as has been observed in vacuum GR [65] and in charged cases [66]. This spurious radiation decays as the system relaxes to quasi-equilibrium. We adopt maximal slicing conditions, $K = 0$. Using a conformally rescaled extrinsic curvature, we can express it in terms of a vector V^i as

$$\tilde{A}^{ij} = 2\delta^{ik}\delta^{jh}V_{(h,k)} - \frac{2}{3}\delta^{ij}\partial_k V^k. \quad (\text{H.27})$$

Under these assumptions, the problem reduces to determining the vector V^i and the conformal factor ψ to fully specify γ_{ij} and $\tilde{A}_{ij}$. The constraint equations take the form

$$\nabla^2\psi + \frac{1}{8}\psi^{-7}\tilde{A}_{ij}\tilde{A}^{ij} + 2\pi\psi^5\rho = 0, \quad (\text{H.28})$$

$$(\nabla^2 V)^i + \frac{1}{3}\delta^{ij}\partial_j(\partial_k V^k) - 8\pi J^i = 0. \quad (\text{H.29})$$

We also impose the Gauss and no-monopole constraints. Following [53, 57], we rescale the electromagnetic fields as

$$\tilde{E}^i = \psi^6 E^i, \quad \tilde{B}^i = \psi^6 B^i, \quad (\text{H.30})$$

which makes the Maxwell constraints

$$\partial_i \tilde{E}^i = 0, \quad \partial_i \tilde{B}^i = 0, \quad (\text{H.31})$$

independent of ψ and therefore solvable separately from the spacetime variables. Since the equations are linear, solutions may be superimposed. Following [58], we neglect contributions from the magnetic field.

The momentum constraint is linear and does not depend on ψ , allowing us to decompose the solution into vacuum GR and matter contributions:

$$V^i = V_{\text{GR}}^i + V_{\text{M}}^i. \quad (\text{H.32})$$

For V_{GR}^i , we use the standard Bowen–York solutions [67]. For the matter contribution, we solve

$$\nabla^2 V^i + \frac{1}{3} \delta^{ij} \partial_j (\partial_k V^k) - 8\pi J^i = 0, \quad (\text{H.33})$$

where

$$J^i = 2\Pi D^i \phi + 2e^{-2\alpha_0 \phi} \epsilon^{ijk} E_j B_k + \frac{1}{2} e^{4\alpha_1 \phi} \Xi D^i \kappa. \quad (\text{H.34})$$

Assuming maximal slicing, time symmetry, and neglecting contributions from the magnetic field and axion, we obtain $J^i = 0$. Thus $V_{\text{M}}^i = 0$, and the momentum constraints reduce to the standard Bowen–York form.

The Hamiltonian constraint becomes

$$\nabla^2 \psi + \frac{1}{8} \psi^{-7} \tilde{A}_{ij} \tilde{A}^{ij} + 2\pi \psi^5 \rho = 0. \quad (\text{H.35})$$

Separating the singular part of ψ from the finite correction u yields

$$\psi = \sqrt{\left(1 + u + \sum_{n=1}^{N_p} \frac{M_n}{2R_n}\right)^2 - \left(\sum_{n=1}^{N_p} \frac{Q_n}{2R_n}\right)^2}. \quad (\text{H.36})$$

This corresponds to a Reissner–Nordström black hole plus corrections. Following [58], we define

$$\eta = \sum_{n=1}^{N_p} \frac{M_n}{2R_n}, \quad \xi = \sum_{n=1}^{N_p} \frac{Q_n}{2R_n}, \quad \kappa = 1 + u + \eta, \quad (\text{H.37})$$

so that the conformal factor can be written as

$$\psi = \sqrt{\eta^2 - \xi^2}. \quad (\text{H.38})$$

Rearranging the Hamiltonian constraint gives

$$\begin{aligned} \nabla^2 u = -\frac{1}{\kappa} & \left(\partial_a \kappa \partial^a \kappa - \partial_a \xi \partial^a \xi - \partial_a \psi \partial^a \psi \right. \\ & \left. + \frac{1}{8} \psi^{-6} \tilde{A}_{ij} \tilde{A}^{ij} + 2\pi \psi^5 \rho \right), \quad (\text{H.39}) \end{aligned}$$

with

$$\begin{aligned} \rho \equiv & \Pi^2 + (D\phi)^2 + e^{-2\alpha_0\phi} (E_a E^a + B_a B^a) \\ & + \frac{1}{4} e^{4\alpha_1\phi} (\Xi^2 + (D\kappa)^2). \end{aligned} \quad (\text{H.40})$$

Under the same simplifying assumptions as before, ρ reduces to

$$\rho \rightarrow (D\phi)^2 + e^{-2\alpha_0\phi} (E_a E^a). \quad (\text{H.41})$$

This follows [58] closely, except that we include dilaton contributions in ρ . Specifically, we employ a Reissner–Nordström electric field for the electric component and add a dilaton field for a single, spinless black hole. The axion is initialized to zero and allowed to relax to a quasi-equilibrium state during evolution.

The electric and dilaton fields are defined as:

$$M_i \equiv \text{physical mass of puncture } i, \quad (\text{H.42})$$

$$q_i \equiv \text{electric charge of puncture } i, \quad (\text{H.43})$$

$$m_i \equiv M_i - \frac{q_i^2}{2M_i}, \quad (\text{H.44})$$

$$\alpha_i \equiv \tanh^{-1} \left(\frac{q_i}{M_i \sqrt{2}} \right), \quad (\text{H.45})$$

$$f_i(r_i) \equiv \left(r_i + \frac{1}{2} m_i \right)^2, \quad (\text{H.46})$$

$$\bar{\rho}_i^2(r_i) \equiv f_i(r_i)^2 + 2m_i f_i(r_i) r_i \sinh^2(\alpha_i), \quad (\text{H.47})$$

$$\Phi_i(r_i) \equiv -\frac{1}{2} \ln \left(\frac{\bar{\rho}_i^2(r_i)}{f_i(r_i)^2} \right), \quad (\text{H.48})$$

$$\Phi(r) \equiv \sum_{i=1}^{N_p} \Phi_i(r_i). \quad (\text{H.49})$$

and

$$E^i = \psi^{-6} \sum_{i=1}^{N_p} \frac{q_i}{r_i}. \quad (\text{H.50})$$

Perhaps future approaches making use of hyperbolic constraint formulations, such as those proposed in [68–71], could be adapted to produce improved initial data in the future.

H.3 Wave Extractions

The gravitational wave signals are given by the Newman–Penrose scalar derived from the Weyl tensor as:

$$\Psi_4 = C_{\alpha\beta\gamma\delta} k^\alpha \bar{m}^\beta k^\gamma \bar{m}^\delta. \quad (\text{H.51})$$

Here, k and $\bar{m}$ are part of a null tetrad, which we define below. The extraction of the radiated electromagnetic energy is given by two scalar quantities calculated in the Newman–Penrose formalism [56, 72, 73]. The triads are constructed to be orthogonal to the normal unit vector n^ν , and they are defined as:

$$l^\alpha = \frac{1}{\sqrt{2}} (n^\alpha + u^\alpha), \quad (\text{H.52})$$

$$k^\alpha = \frac{1}{\sqrt{2}} (n^\alpha - u^\alpha), \quad (\text{H.53})$$

$$m^\alpha = \frac{1}{\sqrt{2}} (v^\alpha - iw^\alpha). \quad (\text{H.54})$$

Asymptotically, the triad vectors u , v , and w behave as unit radial, polar, and azimuthal vectors $\hat{r}$, $\hat{\theta}$, and $\hat{\phi}$, respectively. Similarly, we can construct analogous quantities for the electromagnetic radiation. With this construction, the two radiative quantities are defined as:

$$\xi_1 \equiv \frac{1}{2} F_{\mu\nu} (l^\mu k^\nu + \bar{m}^\mu m^\nu),$$

$$\xi_2 \equiv F_{\mu\nu} \bar{m}^\mu k^\nu.$$

The radiative terms for the dilaton and axion are defined in a similar way. For the dilaton, we have:

$$\Phi = \Pi + r^i D_i \phi, \quad (\text{H.55})$$

and similarly, for the axion, we have:

$$\kappa = \Xi + r^i D_i \kappa, \quad (\text{H.56})$$

where Π and Ξ are the time derivatives of the dilaton and axion, respectively.

At a given extraction radius R_{ex} , we perform a multipolar decomposition by projecting Ψ_4 , ξ_1 , ξ_2 , Φ , and κ onto spherical harmonics of spin weights $s = -2, 0$, and -1 :

$$\Psi_4(t, \theta, \phi) = \sum_{l,m} \psi^{lm}(t) Y_{lm}^{-2}(\theta, \phi), \quad (\text{H.57})$$

$$\xi_1(t, \theta, \phi) = \sum_{l,m} \xi_1^{lm}(t) Y_{lm}^0(\theta, \phi), \quad (\text{H.58})$$

$$\xi_2(t, \theta, \phi) = \sum_{l,m} \xi_2^{lm}(t) Y_{lm}^{-1}(\theta, \phi), \quad (\text{H.59})$$

$$\Phi(t, \theta, \phi) = \sum_{l,m} \Phi^{lm}(t) Y_{lm}^0(\theta, \phi), \quad (\text{H.60})$$

$$\kappa(t, \theta, \phi) = \sum_{l,m} \kappa^{lm}(t) Y_{lm}^0(\theta, \phi). \quad (\text{H.61})$$

We can calculate the radiated energy from the gravitational, electromagnetic, axion, and dilaton fields as follows [57]:

$$F_{\text{GW}} = \frac{dE_{\text{GW}}}{dt} = \lim_{r \rightarrow \infty} \frac{r^2}{16\pi} \sum_{l,m} \left| \int_{-\infty}^t dt' \psi^{lm}(t') \right|^2, \quad (\text{H.62})$$

$$F_{\text{M}} = \frac{dE_{\text{M}}}{dt} = \lim_{r \rightarrow \infty} \frac{r^2}{4\pi} \sum_{l,m} |\phi_2^{lm}(t)|^2. \quad (\text{H.63})$$

H.4 Table of Tests Done

Binary black hole configurations

(a) Kerr–Sen Same-Sign Charge and Spin-Aligned Black Hole Tests

(χ_1, Q_1)	(χ_2, Q_2)	Dilaton ϕ	Axion κ
$\chi_1 = 0.20, Q_1 \in \{0.50, 0.25, 0.125, 0.0624\}$	$\chi_2 \in \{0.20, 0.10, 0.05\}, Q_2 \in \{0.50, 0.25, 0.125, 0.0624\}$	✓	✓
$\chi_1 = 0.10, Q_1 \in \{0.50, 0.25, 0.125, 0.0624\}$	$\chi_2 \in \{0.10, 0.05\}, Q_2$ as above	✓	✓
$\chi_1 = 0.05, Q_1 \in \{0.50, 0.25, 0.125, 0.0624\}$	$\chi_2 = 0.05, Q_2$ as above	✓	✓

(b) Collisions with Other Black Holes

Black Hole 1 (Q, χ)	Black Hole 2 (Q, χ)	ϕ	κ
Kerr–Sen (0.5, 0.2)	Kerr–Sen (0.5, 0.2)	✓	✓
Kerr–Sen (0.5, 0.2)	Kerr (0.0, 0.2)	✓	✓
Kerr–Sen (0.5, 0.2)	Kerr–Sen (−0.5, 0.2)	✗	✗
Kerr–Sen (0.5, −0.2)	Kerr–Sen (0.5, 0.2)	✓	✗
RN (0.5, 0.0)	Kerr (0.0, 0.2)	✓	✓
KN (0.5, 0.2) [†]	KN (0.5, 0.2) [†]	✓	✓

Appendix I

Waveform Catalog

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